SUPERMAN SUIT

A Photovoltaic Neuronmuscular Safety Net for Off-World Habitats

Photovoltaic textile zones.



Flexible, high-efficiency photovoltaic textile integrated in torso, shoulders, upper arms, thighs, and back. Harvests incident light across spectrum.

Power management + storage



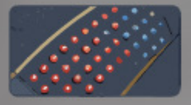
Detachable power management unit with MPPT, charge regulation, protections. and solid-state storage. Mounted at waist/side for balance and access

Distributed sensing

- Thermal
- Humidity
- Pressure
- Gas / Air quality
- Motion / IMU
- Physiological (PPG)

Embedded sensors monitor environmental conditions, suit status, and user state. Data supports safety limits, context-aware assistaone, and graceful degradation

NMES electrode arrays



Detachable power management unit with MPPT, charge regulation, protections. and solid-state storage. Mounted at waist/side for balance and access

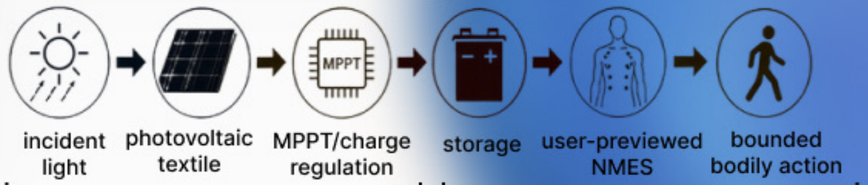
Physical emergency stop



Large, tactile, always accessible button, Immediately disables NMES output and enters safe state



Resource-to-body provenance



user preview affirmative initiation



preview
user reviews proposed assistance (intensity, groups. duration, limits)

affirmative initiation
user deliberately confirms to enable NMES

NMES delivery within limits
system delivers bounded NMES per approved plan

bounded assistance
supports intended movement within safety boundaries

user preview affirmative initiation



monitor
continuous monitoring of power, sensors, and user state

adapt
user deliberately confirms to enable NMES

prioritize
preserve essential functions and safety margin

inform
notify user and log log status: remain transparent